

ROCHELLE, IL, USA

WMO: 722182

Lat: 41.893N

Lon: 89.078W

Elev: 238

StdP: 98.50

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04890

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.5	-19.1	-26.1	0.4	-22.2	-22.7	0.5	-18.2	12.8	-3.7	11.9	-4.1	2.7	270	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	32.2	23.0	30.7	22.8	29.0	22.0	25.5	29.6	24.4	28.4	23.4	27.3	5.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.6	28.1	23.0	18.2	26.8	22.3	17.5	26.1	79.6	29.5	75.0	28.3	71.0	27.2	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.6	9.0	DB	-28.0	34.1	5.3	1.4	-31.8	35.1	-34.9	36.0	-37.9	36.8	-41.8	37.8	(4)
(5)			WB	-28.2	27.3	5.0	1.1	-31.8	28.0	-34.7	28.7	-37.5	29.3	-41.1	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-6.1	-4.9	2.4	9.3	15.8	20.7	21.8	20.6	17.5	10.5	3.0	-3.6
	DBStd	11.34	7.05	6.85	6.08	5.01	5.14	3.37	3.14	2.99	4.26	5.31	5.44	6.31
	HDD10.0	1947	501	419	248	73	8	0	0	0	1	59	217	422
	HDD18.3	3777	759	652	495	273	112	14	6	13	65	251	459	679
	CDD10.0	1578	0	1	12	52	189	321	366	328	227	74	7	1
	CDD18.3	366	0	0	1	2	34	86	114	82	41	7	0	0
	CDH23.3	3452	0	0	10	41	393	792	1049	694	396	77	0	0
	CDH26.7	997	0	0	1	3	134	249	316	163	112	17	0	0
Wind	WSAvg	3.9	4.6	4.6	4.7	5.2	4.6	3.6	2.5	2.2	2.7	3.6	4.4	4.4
Precipitation	PrecAvg	893	47	39	58	82	116	112	92	96	76	59	60	43
	PrecMax	1078	89	107	120	154	308	248	194	217	156	169	130	82
	PrecMin	574	13	13	22	34	17	21	27	1	16	8	5	9
	PrecStd	133	20	25	30	36	71	64	45	46	44	41	38	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	15.1	24.8	27.0	33.2	33.2	33.8	32.4	32.4	28.9	20.3	14.7
		MCWB	9.8	11.7	17.8	15.9	21.2	22.7	26.2	23.7	22.8	20.9	13.8	13.4
	2%	DB	7.6	9.4	18.0	24.2	30.2	31.7	31.7	30.1	30.1	25.4	16.3	10.2
		MCWB	6.4	7.4	13.6	15.0	19.7	22.3	25.2	23.6	21.8	17.6	12.0	8.6
	5%	DB	4.1	5.3	14.7	21.6	27.6	29.8	30.0	28.3	27.5	22.4	13.7	7.0
		MCWB	2.9	3.8	11.0	13.9	18.2	21.8	23.8	22.7	20.0	15.9	10.8	5.9
	10%	DB	2.4	2.9	11.2	18.8	25.1	27.6	28.1	27.2	25.6	19.7	11.3	4.2
		MCWB	1.6	1.7	8.1	12.7	17.7	20.4	22.5	21.6	19.1	15.7	8.4	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	11.7	18.0	18.7	23.2	26.2	27.9	26.1	24.8	22.2	15.7	13.5
		MCDB	10.9	14.3	24.1	22.5	28.5	30.6	32.4	30.1	29.2	27.7	18.1	14.5
	2%	WB	6.6	7.4	14.7	16.6	21.7	24.1	26.1	24.7	23.3	19.8	13.2	8.4
		MCDB	7.5	9.2	17.7	21.3	27.4	28.6	30.1	28.6	27.6	22.6	15.9	9.2
	5%	WB	3.1	4.1	11.6	15.1	20.2	23.0	24.9	23.7	21.9	17.8	10.9	5.6
		MCDB	3.9	5.2	13.9	20.0	25.3	27.9	28.9	27.2	25.9	20.3	13.4	6.7
	10%	WB	1.6	1.9	8.2	13.2	18.9	21.9	23.8	22.6	20.6	15.6	8.7	3.3
		MCDB	2.3	2.9	11.0	18.4	23.5	26.0	27.3	25.8	24.0	19.1	11.1	4.2
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.8	9.5	12.2	12.2	11.3	11.3	11.5	13.0	12.0	9.5	7.8
		MCDBR	9.5	10.7	14.0	16.8	15.2	13.5	12.5	12.6	14.5	15.6	13.3	10.4
		MCWBR	8.2	8.4	9.4	9.5	7.0	6.6	7.0	7.2	7.2	8.5	9.4	9.1
	5% WB	MCDBR	9.0	10.2	12.8	13.9	12.4	11.1	11.2	11.0	11.8	11.9	12.1	9.9
MCWBR		8.0	8.3	9.2	9.1	6.8	6.4	7.0	7.0	6.6	8.4	9.3	9.0	
Clear-Sky Solar Irradiance	taub	0.286	0.304	0.335	0.386	0.417	0.425	0.440	0.421	0.394	0.342	0.302	0.283	
	taud	2.410	2.373	2.398	2.265	2.232	2.256	2.201	2.251	2.301	2.450	2.506	2.445	
	Ebn at Noon	866	903	913	883	860	852	833	838	836	849	845	839	
	Edh at Noon	78	95	105	130	138	135	141	130	115	88	71	70	
All-Sky Solar Radiation	RadAvg	1.70	2.65	3.66	4.65	5.39	6.11	6.08	5.36	4.48	2.94	1.91	1.40	
	RadStd	0.20	0.33	0.32	0.39	0.39	0.46	0.35	0.32	0.38	0.35	0.16	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page